

Video Slides W1: Thinking like an Economist

ECO 101: Principles of Microeconomics

Prof. Freitas, University of Toronto

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Decision Making Overview

Topic: Thinking like an Economist

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Video Learning Objectives

- Introduce the three general principles of individual decision making we will use.

NOTE: Look for these principles in every topic.

Why Analyze Choices?

- Resources are finite.
 - time, money, land, skills, technology, labor, natural resources, personal relationships etc.
- Economics: Broad principles on how to **allocate** these finite resources to their “best” possible use.

Using Models in Economics

- Models: Simplify complex reality, focus on important aspects.
 - We do this by making **assumptions**.
- Use models to make predictions, both quantitative and qualitative.
 - Do the predictions match data? If not, revisit assumptions.

Positive vs. Normative Analysis

Use models to answer different types of questions

- Positive Analysis: Descriptive statements that can be proven true/false.
- Normative Analysis: Makes a value judgment (i.e. should).

Positive vs. Normative Analysis

- Providing academic scholarships to low-income students will increase high school completion rates.
- The government must give \$1000 scholarships to low-income students.

Basic Assumption

- Choose action if $\text{Benefit} \geq \text{Cost}$.
 - We need to correctly evaluate costs and benefits. These may not be obvious or easily measurable.

Quantifying Benefits?

- We need to quantify benefits in a way that makes it easy to compare to costs.
- Ask people to tell us their valuation expressed in \$s ?
 - emotions, bank balance, deadlines...?

Costs vs. Benefits

Deeper hidden assumptions

- People know values with certainty and they are able and willing to make these calculations.
- People maximize well-being (self-interested?)

Principle 1: Always Opportunity Costs

- A choice is not made in isolation, what are the alternative options?
 - i.e. Cost = Opportunity Cost
- All costs are Opportunity Costs

Principle 2: Quantity First

- Quantity choice made first if more than one unit is an option.
 - e.g. IF I were to buy granola bars, how many would I buy?
- Use thinking at the margin to determine quantity choice, IF you were to buy/sell.

Principle 3: Given Quantity, then choose action or not

- Given the quantity I would choose, then decide do I want to buy/sell this quantity or not?
- Given quantity, compare Total Benefits vs. Total Costs for this quantity to determine if you actually want to follow through.

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Summary

- Principle 1: All Costs are Opportunity Costs.
- Principle 2: Quantity choice first based on marginal costs and benefits
- Principle 3: Given quantity choice, use total costs and benefits to decide if you want to take action.

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Opportunity Cost

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Video Learning Objectives

- Explain why economists use Opportunity Costs (OC).
- Explain how to calculate OC.
- Examine factors that could change the OC.

Free Dinner

- You win a free dinner to your favorite restaurant. Use it?
- Check the fine print: can only be used on Nov 14, 7-9 pm.
- You have tickets to see Taylor Swift at that exact time.

Why Opportunity Costs?

- Scarce resources + many opportunities $\Rightarrow$ how to allocate resources?
- Choosing to allocate resource to one option means giving up another option.
- $\Rightarrow$ we need to include forgone opportunity into the cost of taking an action.

Opportunity Costs

- The Opportunity Cost of an action is the **net value** (expressed in \$s) of the **next best** alternative.
- Ignore definition, focus on what it means. Adapt the general principle to your setting.

Information Needed

OC: Opportunity Cost

To calculate the OC for action A, we need information on

- 1) Resources needed for A
- 2) The next best use of these resources, say option B.
- 3) Option B: Resources? Benefit from B?

Opportunity Cost of Action A

OC: Opportunity Cost

- Action B is the next best use of resources used in A
- Everything should be expressed in the common measure of \$s.
- $OC_A =$

Calculating Opportunity Costs

OC: Opportunity Cost

- Idea of OC is intuitive but it can be tricky because it mixes in benefits and costs, and potentially different amount of resources

Illustrate using simple examples.

Calculating Opportunity Costs

OC: Opportunity Cost

- Scenario: you are at Indigo choosing a book to buy.
- Choices: Pride and Prejudice (P&P), Grapes of Wrath (GW) and A Suitable Boy (SB)

Same resources, different benefits

Pride and Prejudice (P&P), Grapes of Wrath (GW), Suitable Boy (SB)

Assume only resource is cash. Could there be other resources I've not accounted for?

Book	Price	\$Benefit	Opportunity Cost
P&P	\$20	\$40	
GW	\$20	\$25	
SB	\$20	\$35	

Different resources, same benefits

Pride and Prejudice (P&P) and A Suitable Boy (SB)

Assume only resource is cash.

Book	Price	\$Benefit	Opportunity Cost
P&P	\$20	\$40	
SB	\$15	\$40	

No price but different benefits

Pride and Prejudice (P&P) and A Suitable Boy (SB)

Assume each book takes time to read

Book	Price	Hours	\$Benefit	Opportunity Cost
P&P	\$0	5 hrs	\$40	
SB	\$0	15 hours	\$30	

Same benefits, different prices

Pride and Prejudice (P&P) and A Suitable Boy (SB)

Assume only resource is cash.

Book	Price	\$Benefit	Opportunity Cost
P&P	\$20	\$40	
SB	\$0	\$40	

Opportunity Cost

Pride and Prejudice (P&P) and A Suitable Boy (SB)

- $OC_{P\&P} =$

Changing Opportunity Cost

Pride and Prejudice (P&P) and A Suitable Boy (SB)

- $OC_{P\&P} = \$p_{P\&P} + \$time_{P\&P} + \$Value_{SB} - \$p_{SB} - \$time_{SB}$

Examples

- $\$time_{P\&P} \uparrow \Rightarrow OC_{P\&P}$
- $\$time_{SB} \uparrow \Rightarrow OC_{P\&P}$

Lessons

OC: Opportunity Cost

- OC includes (some) \$ price. E.g. extra \$ that would not have to be spent on next best alternative.
 - But: Not all \$ prices are OC (e.g. prices of next best alternative is the same)

Lessons

OC: Opportunity Cost

- OC can include costs not visible to an accountant. E.g. \$ value of benefits from other alternative.
 - But: not all implicit costs seen in OC (e.g. benefit from the next best alternative is the same.)

Twist

OC: Opportunity Cost

- Missing from our discussion so far...quantity
- Lunch costs \$15. Next best use is 3 cups of coffee.
Which cups do I give up?
- If you have to give up multiple units of the next best alternative, give up the lowest value ones first.

Summary

OC: Opportunity Cost

- $OC_A = Resources_A + Value_B - Resources_B$
- To understand how OC changes, focus on how the elements of the OC calculate change.

Economic Costs

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- Define: implicit and explicit costs, sunk costs.
- Explain the difference between economic and accounting costs.
- What does zero (economic) profits mean?

Explicit vs. Implicit Costs

Economic costs = Opportunity Costs (OC)

- Explicit costs: costs that you actually pay (think price paid)
- Implicit costs: costs that are relevant to economic decision making but not something you can get a receipt for. e.g. time spent reading book.

Economic vs Accounting Costs

- Opportunity Costs = Explicit + Implicit Costs
 - Economic profits = Revenue – Opportunity Costs
- Accounting Costs = Explicit costs.
 - Accounting profits = Revenue – Accounting Costs

Example

- You paid \$150 entry to club, drinks free. It is 1am and you have a midterm tomorrow.
- Friend says you need to get your money's worth and get another drink. Should you?

Sunk Costs

- Sunk costs are costs that have already been spent (or committed to) and you cannot recover (or get out of paying).
- They should not affect choices.
- If you cannot reallocate the resource, it can't be used anywhere else and so has no impact on opportunity cost.

Zero (Economic) Profits

Zero Economic Profits

- Economic Profits = Revenues - Opportunity Costs
- Zero Economic Profits $\Rightarrow$ Revenues = Opportunity Costs
- You are making as much as your next best alternative!

Summary

- Pay attention to the types of costs: sunk, implicit vs. explicit.
- Economists focus on Opportunity Costs.
- Use concepts to calculate Opportunity Cost, not explicit + implicit.

Willingness to Pay

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Video Learning Objectives

- Define: Willingness to Pay (WTP)
- Explain why we use it and how to calculate it.
- Discuss factors that could change WTP.

Revisit Opportunity Cost (OC)

- OC relevant for decisions.
- Problem some parts may not be observable (e.g. implicit costs) and that makes it hard to use.
- Q: Can we move things around to make it easier to use?

Willingness to Pay (WTP)

- $\text{Benefits (in \$)} \geq \text{Opportunity Costs(OC)}$
- Recall: $\text{OC} = \text{Explicit Costs} + \text{Implicit Costs}$.
- $\text{Benefits} - \text{Implicit Costs} \geq \text{Explicit Costs}$
- Define: WTP as $\text{Benefits} - \text{Implicit Costs}$
- Recall: Explicit Costs usually Price.

$$WTP \geq \text{Price}$$

Example

Pride and Prejudice (P&P), Suitable Boy (SB)

How much am I willing to pay for P&P?

Book	Price	\$Benefit	Opportunity Cost
P&P	\$20	\$40	
SB	\$20	\$35	

Decision

Pride and Prejudice (P&P) and A Suitable Boy (SB)

- Buy P&P if $\$Value_{P\&P} \geq OC_{P\&P}$
- $OC_{P\&P} = \$p_{P\&P} + \$time_{P\&P} + \$Value_{SB} - \$p_{SB} - \$time_{SB}$

Willingness to Pay

Pride and Prejudice (P&P) and A Suitable Boy (SB)

Buy P&P if

$$\$Value_{P\&P} \geq \$p_{P\&P} + \$time_{P\&P} + \$Value_{SB} - \$p_{SB} - \$time_{SB}$$

Changing Willingness to Pay

Pride and Prejudice (P&P) and A Suitable Boy (SB)

- $WTP_{P\&P} =$
 $\$Value_{P\&P} - \$time_{P\&P} - \$Value_{SB} + \$p_{SB} + \$time_{SB}$

Examples

- $\$Value_{SB} \uparrow \Rightarrow WTP_{P\&P}$
- $\$price_{SB} \uparrow \Rightarrow WTP_{P\&P}$

Summary

WTP: Willingness to Pay, OC: Opportunity Cost.

- Buy if Benefits $\geq$ OC can be rewritten as
buy if WTP $\geq$ Price.
- Intuitive and easy to compare to price!
- Especially useful for thinking at the margin.

Thinking at the Margin

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Video Learning Objectives

- Explaining the difference between Marginal and Total Costs (Benefits).
- Calculate Marginal Costs (and Benefits).
- Decision making with a quantity dimension.

Note: For Marginal calculations, it is much easier to convert the Benefits vs. Opp. Cost equation into a Willingness to Pay vs. Price equation first.

Decisions

- Should I buy granola bars?

We approach it by working backwards...

- 1) If I buy granola bars how many will I buy?
- 2) Buy this quantity or not?

How Many?

Overview

- Examples: how many workers to hire, hours to study etc.

Process

- 1) Break into series of smaller decisions
- 2) For the extra unit (bit) compare extra cost to extra benefit.

Why Thinking at the Margin?

MC: Marginal Cost, MB: Marginal Benefit

- Get surplus on the extra unit if extra benefit $\geq$ extra cost.

Note:

- Marginal = Tiny bit extra
- Will use discrete units (i.e. 1 extra unit) for intuition.

Marginal Costs and Benefits

MC: Marginal Cost, MB: Marginal Benefit

You have $N - 1$ units and are considering buying 1 extra unit which will get you to N units

- MC: Increase in costs when purchase the extra unit.

$$MC(N) = \text{Cost}(N) - \text{Cost}(N - 1)$$

- MB: Increase in benefits when purchase the extra unit.

$$MB(N) = \text{Benefit}(N) - \text{Benefit}(N - 1)$$

Marginal Costs and Benefits

MC: Marginal Cost, MB: Marginal Benefit

Process continued

- 3) Keep buying extra unit as long as $MB \geq MC$.
- 4) Stop at the *largest* quantity where $MB \geq MC$
 - Caveat: watch out for weird MCs or MBs

Marginal Costs and Benefits

MC: Marginal Cost, MB: Marginal Benefit



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Example

Price = \$10 per unit. 20% discount if you buy 3 or more.

Quantity	1	2	3	4
Total Benefit	\$11	\$20	\$27	\$32
Marginal Benefit				
Marginal Cost				
Total Cost				
Total Benefit-Total Cost				

How many units do you buy?

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Why Thinking at the Margin?

MC: Marginal Cost, MB: Marginal Benefit

- Recall: MB vs MC determines quantity IF you were to buy.
- Still need to take a step back and decide if you're going to buy or not and then Total Benefit vs Total Cost becomes important.

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Marginal vs. Total thinking

MC: Marginal Cost, MB: Marginal Benefit

Marginal thinking: add extra unit if $MB \geq MC$.

- Q: Total benefit from 2 hours spent on Netflix is \$50. Opportunity cost of each hour is \$15. Watch for 2 hours?

Changing Quantity Choices

- Things that change Marginal Benefits and/or Marginal Costs change the quantity choices.

Examples

- (Marginal) Value from good changes.
- (Marginal) Cost of good changes.
- (Marginal) Value from next best alternative changes.

Summary

MC: Marginal Cost, MB: Marginal Benefit

- We use thinking at the margin to determine quantity choices.
- Stop at the *largest* quantity where $MB \geq MC$.
- We decide whether to buy this quantity or not based on Total Costs and Benefits.